

Abu Bakar Yasir

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Summary

Full-Stack & AI Engineer with **1+ year of professional experience** shipping production-grade web applications and AI systems end-to-end. Builds multi-tenant SaaS platforms, Retrieval-Augmented Generation (RAG) pipelines, and multi-agent LLM orchestration with Next.js, FastAPI, and Django REST Framework. Strong in REST API design, secure authentication (JWT/RBAC), vector databases, and cloud deployment.

Education

BS Computer Engineering, COMSATS University Islamabad (CUI) — *Sep 2022 – Jun 2026* CGPA: 3.29 / 4.0
• Lahore, Pakistan

Work Experience

Full-Stack & AI Engineer — [Spiral Lab](#) · *April 2025 – Present*

- Shipped **production-grade, full-stack applications** end-to-end across 3+ products using Next.js, FastAPI, Django, Node.js, and Supabase/PostgreSQL, owning backend, AI layers, async pipelines, and frontend.
 - Built **AI-driven systems** including Retrieval-Augmented Generation (RAG) pipelines and multi-agent LLM orchestration (OpenAI, Anthropic Claude, Groq), integrating vector databases (Qdrant, ChromaDB) for low-latency semantic retrieval.
 - Engineered **secure, multi-tenant architectures** with JWT authentication, Role-Based Access Control (RBAC), row-level security, rate limiting, and audit logging, plus async pipelines (Django-Q2, Redis, WebSockets) deployed to Railway and Vercel with CI auto-deploy.
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Projects

Magnai — Vertical AI SaaS for UK Public Affairs · *Live*

Python, Django REST Framework, PostgreSQL, Anthropic Claude Agent SDK, OpenAI, Django-Q2, Redis, Railway

- Built and deployed a **multi-tenant Django SaaS** with a multi-agent AI system (orchestrator plus parliament/media/Twitter sub-agents, each with its own tool registry) monitoring UK, Scottish, and Welsh data to auto-generate AI-authored intelligence reports.
- Developed **robust async monitoring pipelines** (Django-Q2 and Redis) with race-condition protection, a security layer (rate limiting, anomaly detection, CSP), and a stakeholder-mapping engine with relevance scoring.

Lynxflow-Health — Healthcare Management System (SaaS) · *Live*

Next.js, Node.js, Supabase (PostgreSQL), REST APIs

- Built a **responsive, role-based healthcare platform** (Clinical and Super Admin) with row-level security (RLS) and HIPAA-aligned data handling; integrated Tenovi smart-pillbox hardware via webhooks for

medication-adherence tracking, plus in-platform device ordering and a vulnerability-scanning bot.

neuroMedica — Medical RAG System for Clinical Decision Support · *Live*

Next.js, TypeScript, FastAPI, Qdrant, Neo4j, PubMedBERT, Anthropic Claude, TensorFlow, PyTorch, Docker

- Designed a **hybrid-retrieval RAG pipeline** fusing dense (PubMedBERT/Qdrant), sparse (BM25), and knowledge-graph (Neo4j/UMLS) retrieval via Reciprocal Rank Fusion, with a CRAG self-evaluation loop to cut hallucination for clinical safety.
- Built **deep-learning diagnostic modules**: a Keras CNN for ECG arrhythmia classification (R-peak detection via neurokit2) and a DenseNet-121 (PyTorch) multi-label chest X-ray pathology classifier; deployed full stack via Docker on Railway.

Slack-Bounty-Bot — Real-Time Replit Bounty Notifier · *GitHub*

Python, Flask, Firecrawl API, Slack Webhooks, Vercel

- Built a **Python microservice** that scrapes and filters high-value Replit bounties in real time and pushes automated Slack alerts; deployed as a Flask API on Vercel with CRON-based continuous monitoring.

Skills

- **Languages:** Python, JavaScript, TypeScript, C++, SQL
- **AI / ML:** RAG, LLM Integration (OpenAI, Anthropic Claude, Groq, Gemini), Multi-Agent Orchestration, Embeddings, Prompt Engineering, TensorFlow, PyTorch
- **Backend:** FastAPI, Django, Django REST Framework, Node.js, Express.js, REST APIs, WebSockets
- **Frontend:** Next.js, React.js, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, Supabase, ChromaDB, Qdrant, Neo4j
- **DevOps & Tools:** Docker, Railway, Vercel, Redis, Git, GitHub, Postman